DISTANCE Chapter 02

The Big Bang and the Diffusion of the Universe

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1 Before Expansion — A Universe of Extremely Close Relationships

When we imagine the beginning of the universe, we often picture a familiar scene.

There is an extremely small region of space. Within it, an enormous amount of energy and matter is compressed. At some point, it begins to expand explosively. Space grows, matter moves farther apart, stars and galaxies gradually form, and the expansion continues to this day.

This picture is so familiar that we often overlook the fact that it combines different kinds of explanation.

What do we actually observe?

And what have we assumed in order to explain those observations?

To say that the relationships among distant galaxies are changing is not exactly the same as saying that space itself is expanding. Likewise, to say that the early universe was far denser than the universe we observe today is not exactly the same as imagining that everything was compressed inside a tiny region of geometric space of the kind familiar to us now.

DISTANCE does not begin by rejecting conventional cosmology.

Instead, it changes the starting point of the question.

Must we begin by asking about the size of space in order to describe the early universe?

Or can we first ask:

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What kinds of relationships existed among the constituents of the universe at that time?

When the language developed in Chapter 1—Difference, Distance, Momentum, and Equalization—is first applied to the actual universe, this is where the inquiry begins.

Not How Small, but How Strongly Related

The word small creates a powerful image when we think about the early universe.

We naturally imagine a tiny sphere.

Everything that would eventually constitute the universe is packed inside it.

Then the sphere grows until it becomes the universe we observe today.

But this image already contains a particular intuition about geometric space.

Small relative to what?

Where was this small universe located?

What was outside it?

And if space itself cannot yet be assumed to have had the same meaning in the earliest state of the universe as it does in our present descriptions, is it appropriate to imagine the early universe as though it were a small object sitting inside familiar space?

DISTANCE approaches the question differently.

What matters about the early universe need not first be how geometrically small it was.

A more fundamental question may be:

How strongly effective were its relationships?

If countless constituents that now interact only weakly were once involved in extremely strong relationships, then in the language of DISTANCE those relationships can be described as having very short Distances.

Here, short does not mean a small geometric separation.

As defined in Chapter 1:

short Distance = strongly effective relationship

This distinction is crucial to everything that follows in Chapter 2.

Extremely Short Distances

Suppose there is a Difference between two states.

If that Difference is barely effective within their relationship, their Distance is long.

If, on the other hand, the Difference strongly participates in changing the other state, their Distance is short.

When the early universe is redescribed in the language of DISTANCE, we can therefore imagine a state in which countless relationships had extremely short Distances.

A change in one relationship strongly affects many others.

A change in one state alters the conditions of numerous surrounding relationships.

It becomes increasingly difficult to treat any constituent as completely independent of the others.

The defining feature of the early universe, then, need not be expressed only as a high density of matter.

At a more relational level, it may also be read as an:

extreme density of effective relationships

This is not an attempt to replace the conventional physical quantity of matter density with a new measurable quantity.

DISTANCE is not denying the observable quantities used by physics. It is asking whether the relational condition underlying those quantities can also be read in another language.

An extremely dense material state may simultaneously have been a state in which relationships among constituents were effective to a degree difficult to compare with those of the present universe.

It is this relational aspect that DISTANCE brings to the foreground.

A Close Relationship Is Not a Stable Relationship

An important misunderstanding must be prevented here.

When we hear that a relationship has a short Distance, it is easy to interpret this as a stable condition.

The constituents are close.

Because they are close, they are tightly bound.

Because they are tightly bound, the state appears stable.

And because it is stable, we may expect little change.

But the Momentum–Distance Duality established in Chapter 1 reveals another side of the relationship:

shorter Distance ↔ stronger effective Momentum

A short Distance is not a state in which change has already ended.

It is a state in which strong Momentum remains effective.

If the early universe is understood as a relational network characterized by extremely short Distances, it therefore does not represent a perfectly stable universe.

Quite the opposite.

Such a network contains enormous unresolved Momentum.

The extreme relational density of the early universe is therefore associated with an extreme capacity for change.

We may express this conceptually as:

extreme relational density → extremely short effective Distances → enormous unresolved Momentum

This is the first conceptual shift in the DISTANCE interpretation of the early universe.

Concentration of Energy and Concentration of Relationships

Conventional descriptions characterize the early universe as an extremely high-energy state.

DISTANCE does not reject this.

It asks whether high energy can also be read relationally.

The concentration of energy need not be imagined only as a large quantity of something stored inside a container.

It may also describe a state in which countless Differences are strongly effective,

in which those Differences are deeply interconnected,

and in which a change in one relationship alters many others.

From this perspective, the high-energy state of the early universe is not independent of its extremely dense relational condition.

There is an Energy Gap.

That Gap becomes strongly effective within a relationship.

Distance is short.

Strong Momentum is therefore effective.

And each Momentum coexists with countless others.

The early universe becomes difficult to imagine as merely a collection of independent particles, each moving with its own independent momentum.

It is better approached as an extremely coupled relational state in which countless Momentum continually condition one another.

There Was Not One Enormous Momentum

The phrase enormous Momentum also requires care.

It does not mean that the entire early universe possessed one gigantic Momentum pointing in a single direction.

In DISTANCE, Momentum is relational directionality.

Where there are countless relationships, countless Momentum coexist.

Some reinforce one another.

Some redirect one another.

Some constrain the expression of others.

And even a small change in relational conditions may alter which Momentum becomes effective.

The early universe should therefore not be represented as one enormous arrow.

It is more appropriately understood as a state in which countless extremely strong Momentum coexisted.

This means that the subsequent evolution of the universe need not be interpreted only as an event in which a single force pushed everything outward.

It may instead have been a process in which countless relationships changed simultaneously and the relational network itself was continually reconfigured.

This provides the basis for interpreting the Big Bang as diffusion in the next section.

Compressed Matter, or Compressed Relationships?

We often describe the early universe as a compressed universe.

But this expression can be read in two different ways.

The first is geometric compression.

A great deal of matter occupies a small geometric volume.

The second is relational compression.

Countless states participate in strongly effective relationships and are connected through very short relational Distances.

These two descriptions need not be mutually exclusive.

DISTANCE, however, explores whether the second can serve as the more fundamental starting point of explanation.

Doing so allows us to describe the extremity of the early universe without first having to define the size of space itself.

We can begin by saying:

The early universe was an extremely compressed relational network.

Only afterward do we ask how that relational condition corresponds to the geometric configuration we observe or reconstruct.

The order of explanation is reversed.

Our conventional intuition often proceeds as follows:

small space → high density → strong interaction

DISTANCE asks whether we can instead begin with:

strongly effective relationships → short relational Distances → extreme relational density → observed geometric configuration

There is no need at this stage to decide which of these is ultimately more fundamental.

What matters is that the latter possibility should not be excluded from the beginning.

Was There an "Outside" of the Early Universe?

Once we imagine a small universe exploding, we almost automatically imagine an outside.

Matter explodes outward from a center.

But when we are talking about the universe as a whole, this picture immediately becomes problematic.

If the universe did not explode inside some larger external space, what does outward mean?

A relational approach allows DISTANCE to formulate the problem differently.

Equalization does not require a pre-existing geometric outward direction.

There are Differences among relationships.

Those Differences become effective through different Distances.

Countless Momentum emerge within those relationships.

And the resolution of those Momentum changes the relational network itself.

The transformation of the early universe therefore need not begin with a single geometric direction:

center → outside.

At a more fundamental level, it may instead be described as:

strongly coupled relationships → Momentum resolution → relational rearrangement

The macroscopic result may appear observationally as separation and diffusion.

The distinction is important.

Diffusion does not conceptually require an external space into which the universe must expand.

What it requires, in the DISTANCE description, is Difference, relationships through which Difference becomes effective, and Momentum expressed through those relationships.

When Ink Diffuses Through Water

To make this distinction more intuitive, consider a drop of ink placed in water.

At first, the ink is concentrated within a small region.

As time passes, it spreads throughout the water.

We call this diffusion.

The ink does not spread because the small region of space that initially contained it expands.

Its concentrated configuration is transformed as relationships among its constituents and the surrounding medium change.

Of course, the universe is not ink, and it is not water.

Nor does the analogy imply that some enormous external container existed outside the early universe.

The analogy must therefore not be mistaken for a physical model.

It is useful only for illustrating one conceptual distinction:

The observation that something spreads is not logically identical to the claim that the space in which it exists has itself expanded.

This distinction is why DISTANCE considers the Big Bang from the perspective of diffusion.

The first question is whether cosmic change can be described in terms of changing relationships before the expansion of space itself is taken as the starting premise.

But the Early Universe Was Far More Complex Than a Drop of Ink

The ink analogy has an obvious limitation.

Ink spreads into water that already exists.

A background is already given.

Space is already assumed.

The distinction between ink molecules and water molecules is already established.

None of these conditions can simply be transferred to the early universe.

Cosmic diffusion in DISTANCE is therefore not a process in which matter spreads outward through a pre-existing empty space.

It is instead a rearrangement of the relational network itself.

Rather than beginning with an independent spatial stage, we begin with relationships changing,

Distances changing,

Momentum being resolved,

and a configuration emerging that we subsequently read in spatial terms as separation.

In this sense, cosmic diffusion means something broader than ordinary material diffusion.

It is closer to relational diffusion, or more precisely, to Momentum-resolution-driven relational rearrangement.

Where Was the Boundary of the Early Universe?

Beginning with relationships also changes another familiar question.

How large was the early universe?

Where was its boundary?

What lay beyond it?

These are natural questions when we begin by treating the early universe as a geometric object.

But if the early universe is approached as an extreme relational state, the meaning of boundary itself must be reconsidered.

Must a boundary necessarily be the surface of some geometric sphere?

Or might it also be a relational boundary through which one configuration of relationships becomes distinguishable from another?

We will not answer that question here.

It will return in a more concrete form in Chapter 4 when we examine Islands and physical boundaries.

For now, only one caution is necessary:

We should not automatically treat the early universe as though it were an ordinary object and therefore assume a surface and an outside.

That alone substantially changes the conceptual picture of the Big Bang.

Can Change Be Described Without Putting Space First?

This brings us back to the central question of Chapter 1.

We are not trying to eliminate space.

Nor are we declaring that space is unreal.

We are changing the order of explanation.

The familiar intuitive sequence is roughly:

Space exists. Matter exists within it. Relationships arise among matter. Those relationships produce motion.

DISTANCE tests the reverse possibility:

Difference exists. Difference becomes effective within relationships. Distance is constituted. Momentum appears. Momentum is resolved and the relational network is rearranged. That relational transformation gives rise to configurations that we read spatially.

If such an explanation is possible, space need not always be treated as an absolute stage that must be given before any physical phenomenon can be described.

Spatial structure itself may be one way of reading a relational state.

This is the first step in extending the question raised in Chapter 1 into the history of the universe.

The Early Universe Was Not at Rest

The expression initial state can sometimes suggest a static picture.

We may imagine that at one moment the universe existed in an extremely compressed but motionless state, and that in the next moment an event called the Big Bang began.

DISTANCE gives us no reason to require such a picture.

A relationship with extremely short Distance is a relationship with extremely strong Momentum.

Such a state already contains powerful directionality of change.

There is therefore no need to draw a sharp conceptual boundary between a perfectly motionless state "before" the Big Bang and a moving universe "after" it.

What we call the Big Bang may instead be understood as the observable resolution of extreme relational Momentum that was already present in the relational state itself.

From this perspective, even the question:

"What started the Big Bang?"

changes its form.

There is no need to assume that an external first force acted upon a completely static universe.

Momentum was already present.

As established in Chapter 1:

The existence of Momentum is never the question. Momentum is always present.

The question is not whether Momentum existed.

The question is which Momentum became effective under which relational conditions.

Even the Word "Beginning" Requires Caution

For the same reason, the expression the beginning of the universe should be used carefully.

We may trace the history of the observable universe backward toward an extreme early state.

But we do not need to assume that this necessarily represents the absolute beginning of existence itself.

DISTANCE v1.1 does not extend this discussion into ontology.

Why something exists rather than nothing,

how Difference first arose,

why Momentum exists at all—

these questions lie beyond the scope of this chapter.

The question required in Chapter 2 is much more limited:

Can the extreme early state of the observable universe be reread as a state of extraordinarily strong relationships and unresolved Momentum?

That is enough.

This limitation matters.

DISTANCE is not attempting to replace a gap in cosmology with another creation narrative.

Resolution Rather Than Explosion

The image of the Big Bang can now be rearranged.

The familiar popular picture emphasizes:

something small → explosion → outward expansion.

DISTANCE instead begins with:

extreme relational density → extremely short effective Distances → enormous unresolved Momentum → Equalization → Momentum resolution → increasing relational Distance → relational rearrangement

At this stage, we do not yet need the word expansion.

We have not yet decided what, exactly, is expanding.

What we can say first is that relationships changed.

Distance changed.

Momentum was resolved.

And through that process, the configuration of the universe changed.

At the macroscopic level, that transformation may appear as separation and diffusion.

This is the conceptual starting point of Chapter 2.

From the Early Universe to the Present Universe

The universe we observe today looks radically different from its extreme early state.

Stars are separated by enormous geometric distances.

Galaxies are distinguishable from one another.

Countless Islands exist—planets, rocks, stars, and living systems.

Some relationships are extremely strong.

Others are so weak that they can almost be ignored.

Relational Distances have become extraordinarily diverse.

How, then, did such a differentiated relational network emerge from the extreme relational density of the early universe?

It is not enough simply to say that everything moved farther apart.

Some relationships weakened,

while others became newly strong.

Some Momentum were resolved,

while others became effective.

And structures formed in the process.

The transformation from the early universe to the present universe therefore cannot be described only as dispersion.

It is also a continuing process of rearrangement.

This question will become more concrete as Chapter 2 proceeds.

Before Expansion, There Were Relationships

The central idea of this section can now be compressed into one sentence:

Before asking how the universe expanded, DISTANCE asks how its relationships were configured.

Before asking how small the universe was,

we ask how strongly effective its relationships were.

Before asking what flew outward,

we ask which Momentum were being resolved.

Before asking how much space expanded,

we ask how the relational network changed.

Changing the order of these questions does not automatically invalidate conventional cosmology.

But it allows us to separate observation from the interpretation we have placed upon it.

And within that separation, another way of reading the Big Bang becomes possible.

The Next Step — Big Bang as Diffusion

If the early universe was a relational state characterized by extremely short Distances and enormous unresolved Momentum, the next question follows naturally.

How was that Momentum resolved?

Why did its resolution appear as the vast separation we observe?

And must that separation be interpreted only as space itself expanding?

In the next section, DISTANCE reconsiders the Big Bang not primarily as an explosive event, but as a diffusion process through which an extreme relational condition began to be resolved.

What matters is not the picture of matter flying outward into some external space.

What matters is that relationships changed.

Distance changed.

Momentum was resolved.

And through that process, the configuration of the universe was transformed.

That is the starting point of:

2.2 Big Bang as Diffusion

2 Rereading the Big Bang as Diffusion

In the previous section, we changed the starting point from which we look at the early universe.

Instead of asking first how small the universe was in geometric space, we asked how strongly effective its relationships were.

From that perspective, the early universe was not merely a condition in which a large amount of matter occupied a very small region.

Countless Differences were strongly effective,

their relational Distances were extremely short,

and therefore countless strong Momentum remained unresolved.

We summarized that condition as:

extreme relational density → extremely short effective Distances → enormous unresolved Momentum

The next question follows naturally.

How did such a strongly coupled relational state become the universe we observe today?

The familiar answer is expansion.

DISTANCE asks one more question before accepting that as the starting language.

Must the vast separation and transformation of the universe be described from the outset as the expansion of space itself?

Or can we first describe it as:

the resolution of extremely strong relationships through a continuing rearrangement of the relational network?

From this second perspective, DISTANCE rereads the Big Bang as a process of diffusion.

The Image of an Explosion

The name Big Bang is powerful.

The expression itself invites an image of explosion.

Something concentrated in one place bursts apart.

Matter flies outward from a center.

It spreads through an increasingly large region.

Modern cosmology, of course, does not describe the Big Bang simply as an ordinary explosion occurring inside pre-existing space.

And yet our everyday intuition remains strongly attached to the imagery of explosion.

An explosion has a center.

It has an outward direction.

And it requires a space into which the exploding material can spread.

When the early universe itself is the subject, all three intuitions become problematic.

Was there a single geometric center of the Big Bang?

Can an absolute outward direction be defined from that center?

And can we assume some larger external space into which the universe expanded?

DISTANCE does not solve these problems by adding another background space.

Instead, it returns to relationships as the basic unit of change.

What the early universe requires first is not a center and an outside.

There is Difference.

There are relationships in which those Differences are strongly effective.

There are short Distances.

And there is strong Momentum within those relationships.

Those conditions are already sufficient for change to possess directionality.

Momentum Does Not Wait for an External First Impulse

The image of an explosion raises another question.

What started it?

What was the first impulse?

What caused a previously motionless universe to begin moving?

But from the perspective established in Chapter 1, DISTANCE asks a different question.

Momentum is always present.

Momentum is not merely a quantity later added to an already independent object from outside.

When Difference is effective within a relationship, that relationship already contains directionality of change.

That directionality is Momentum.

If the early universe was a relational network characterized by extremely short Distances, then it already contained enormous Momentum.

There is no need to assume:

first, a perfectly static universe;

then, an external event that created Momentum;

and finally, motion.

The more appropriate question is:

Among the enormous number of coexisting Momentum, which became effective under which relational conditions, and how were they resolved?

Once the question is posed in this way, the Big Bang begins to look less like an external first impulse and more like a vast process through which already existing relational Momentum became effective and began to be resolved.

Where Does Resolution Go?

The moment we use the word resolution, another question appears.

Where does Momentum resolve toward?

Is there a final location?

Does it move away from a center?

Does it move toward some equilibrium point?

DISTANCE does not require a geometric destination to be defined in advance.

Momentum is relational directionality.

Its resolution is therefore first of all a change in relationship.

When Momentum in the A-B relationship is resolved, the relational state of A and B changes.

That change may alter the A-C relationship.

It may alter the B-D relationship.

The resolution of one Momentum is therefore not simply an event in which two objects change position.

It changes the conditions of the relational network as a whole.

This becomes especially important in the early universe, where relationships were extraordinarily dense and strongly coupled.

A change in one relation could alter many others.

Those altered relations would in turn modify the conditions of the original one.

The resolution of the early universe should therefore not be imagined as a single arrow pointing outward.

It is more appropriately understood as a simultaneous reconfiguration of the relational network.

This is the reason DISTANCE uses the language of diffusion.

What Is Diffusing?

Here too, caution is necessary.

In everyday physics, diffusion usually describes something spreading through something else.

A fragrance spreads through air.

Ink spreads through water.

Heat propagates through matter.

Substances of different concentration mix.

All such examples already assume space, a medium, and constituents moving within it.

If cosmic diffusion were understood in exactly the same way, we would return immediately to the same old question:

What did the early universe diffuse into?

What was the surrounding medium?

DISTANCE therefore does not use diffusion to mean:

matter spreading into external space.

What spreads, more fundamentally, is the path of relational resolution.

An extremely concentrated relational configuration changes.

Momentum that had been strongly effective in short-Distance relationships is resolved.

As that Momentum is resolved, the effectiveness of those relationships weakens and their Distances increase.

Those changes reconfigure other relationships.

And this occurs across countless relationships simultaneously.

Cosmic diffusion is therefore better described as:

global relational rearrangement driven by Momentum resolution

rather than as simple material dispersion through pre-existing space.

Diffusion and Increasing Distance

At this point, the directional principle corrected in Chapter 1 must be applied without ambiguity.

We often describe the resolution of Difference by saying that two states “become closer.”

That everyday expression makes Equalization sound like a process in which Distance decreases.

But Distance in DISTANCE does not mean similarity.

It means how strongly a Difference is effective within a relationship.

Thus, if a relation in the early universe had a very short Distance, it carried very strong effective Momentum.

As that Momentum is resolved, the relationship becomes less effective.

Therefore:

Momentum resolution → increasing Distance

This is central to the interpretation of cosmic diffusion.

To reread the Big Bang as diffusion does not mean that the universe sought ever shorter Distance.

It means that the enormous Momentum associated with the extremely short relational Distances of the early universe was resolved, increasing the Distances of those relationships and reconfiguring the network as a whole.

The directional sequence remains:

short Distance → strong Momentum → Equalization → Momentum resolution → longer Distance

Does Everything Simply Move Farther Apart?

An immediate problem follows.

If Distance increases as Momentum is resolved, does the history of the universe simply consist of everything moving endlessly away from everything else?

No.

First, relational Distance and geometric separation are not the same thing.

More importantly, when one Distance increases, the relational conditions of the network change.

That may make another relationship more strongly effective.

Suppose Momentum in A-B is resolved and the states of A and B change.

A relationship between A and C that had previously been weak may then become important.

Its Distance may become relatively short, and a new effective Momentum may emerge.

Cosmic diffusion is therefore not a process in which every relational Distance monotonically increases at the same time.

Rather:

some Distances increase, other relationships become newly short, some Momentum are resolved, and others become effective.

The evolution of the universe is not merely dispersion.

It is continuous relational rearrangement.

This is also what later allows the formation of stars, galaxies, and other physical

Islands to be understood without treating structure formation as the opposite of diffusion.

Returning to the Ink-in-Water Analogy

In the previous section, we introduced the image of a drop of ink diffusing through water.

We can now examine that analogy more carefully.

Immediately after the ink is introduced, its concentration is localized.

There is a strong Difference between the ink-rich region and the surrounding water.

As the ink spreads, that Difference decreases.

The possibility of change that was initially concentrated in one region becomes distributed through a wider set of relationships.

This gives us an intuitive image of cosmic diffusion.

But the decisive difference remains.

Ink has an external environment.

There is already water into which it can spread.

Boundary conditions are already given.

The universe as a whole cannot simply be assumed to possess such an external medium.

What we should take from the analogy is therefore not the picture of molecules traveling into surrounding space.

It is only this:

as concentrated Difference is resolved, the original relational configuration can no longer remain unchanged.

A new configuration emerges.

Cosmic diffusion focuses on this change of configuration.

Can Separation Occur Without Expanding Space?

We can now ask a more difficult question.

If we temporarily suspend the explanation that space itself expands, how can vast separation arise?

The question seems almost impossible if geometric space is assumed to be the most fundamental starting point.

If the geometric distance between A and B increases, then:

A must move,

B must move,

both must move,

or the space between them must expand.

But DISTANCE first descends below this geometric level.

Must A and B themselves be assumed to be permanently independent objects with fixed identities from the beginning?

Or might the structures that we later call A and B themselves emerge through relational rearrangement?

If the relational network of the early universe changes, the structures that later become distinguishable as A and B may emerge through that process.

It may therefore be too simple to project today's identifiable objects backward and imagine the same objects merely packed more tightly into a smaller space.

Cosmic diffusion is not simply:

objects moving apart.

At a deeper relational level, it may involve:

relationships changing, configurations differentiating, Islands emerging, and geometric separation becoming observable.

Relational Resolution Is Not Identical to Geometric Separation

But we must avoid creating a new oversimplification.

We should not write:

relational Distance increases = geometric distance increases.

As established in Chapter 1, these are distinct concepts.

Two objects may become geometrically closer while a particular relational Distance increases.

Conversely, they may become geometrically farther apart while another relationship becomes more strongly effective.

Cosmic separation after the Big Bang must therefore not simply be identified with increasing relational Distance.

The more limited claim DISTANCE can make is:

the extreme relational network of the early universe underwent large-scale Momentum resolution;

this altered relational Distances;

those changes caused extensive relational rearrangement;

and part of that rearrangement may appear at the macroscopic observational level as geometric separation.

Conceptually:

Momentum resolution → relational Distance change → relational rearrangement → possible observable geometric separation

The final step is an observable expression, not the definition of relational Distance itself.

This distinction must remain intact throughout all later physical arguments.

Why Use the Word "Diffusion"?

Why use diffusion at all?

Would change or rearrangement not be sufficient?

The word is useful because it captures the overall direction in which an initially extremely concentrated relational state ceases to remain concentrated.

The early universe was relationally dense.

Many relationships had short Distances.

Strong Momentum was effective across the network.

As those Momentum were resolved, the original extreme relational concentration could no longer be maintained.

Relationships were rearranged into increasingly differentiated configurations.

At the macroscopic level, this resembles spreading.

But what is spreading is not a single body of matter through space.

It is the effective concentration of relationships being redistributed into a wider and more complex relational configuration.

In this limited sense, diffusion fits the DISTANCE perspective better than explosion.

Explosion strongly implies a center, an outside, and a pre-existing space.

Diffusion allows the focus to shift toward transformation of relationships.

Does Diffusion Require One Direction?

An explosion has an outward direction.

Diffusion does not necessarily require a single geometric direction.

Ink spreads in many directions.

Heat does not propagate only along one absolute vector.

Cosmic diffusion can be understood similarly.

There is no need to assume that the early universe possessed one absolute outward vector.

Each relationship contains its own Difference and Momentum.

Countless Momentum are resolved simultaneously.

Their resolutions alter one another's conditions.

The overall process is therefore not one simple arrow, but a vast number of resolution paths continually forming and changing together.

At larger scales, however, a common directionality may still emerge.

The extreme relational concentration of the early universe did not remain unchanged.

In that sense, there is a global direction to the process.

Later, DISTANCE will develop this more carefully through the concepts of global Momentum and global Equalization.

For now, we do not bring those mechanisms forward prematurely.

It is enough to recognize that the evolution of the early universe need not be reduced to the accidental sum of unrelated local events.

Diffusion Is Not the Same as Increasing Disorder

The image of spreading often makes us think of disorder.

Something concentrated becomes dispersed.

Something organized becomes scattered.

Structure disappears.

But DISTANCE does not define cosmic diffusion through an order/disorder framework.

As one configuration is resolved, another may form.

New relationships become effective.

New Momentum emerge.

New Islands may appear.

The later formation of stars and galaxies cannot be understood simply as everything becoming more randomly dispersed.

Defining diffusion as:

order $\rightarrow$ disorder

would therefore create unnecessary problems.

DISTANCE asks a more direct question:

Which Momentum are being resolved, and how is the relational network rearranged as a result?

Order and disorder may be ways of describing that rearrangement at a particular observational scale.

They are not the definition of diffusion itself.

Diffusion Is Not the Opposite of Structure

This leads to an important consequence.

Structures can form while cosmic diffusion continues.

At first, this sounds contradictory.

How can things gather while the universe is diffusing?

How can stars and galaxies form if Distance is increasing?

The contradiction arises only if relational Distance is identified with geometric distance.

During the resolution of a global Momentum, some constituents may become geometrically closer.

They may form new local relationships.

Those relationships may acquire relatively shorter Distances.

New local Momentum may become effective.

And new Islands may emerge.

Thus global diffusion and local aggregation need not oppose one another.

A possible sequence is:

global Momentum resolution $\rightarrow$ relational rearrangement $\rightarrow$ reconstruction of local Difference $\rightarrow$ new local Momentum $\rightarrow$ local structure formation

This becomes increasingly important later in Chapter 2.

Resolving One Relationship Opens Another

Suppose Momentum in the A-B relationship is resolved.

A and B change.

Then A-C changes.

The conditions of C-D may also change indirectly.

A relationship that was previously almost ineffective may now become effective.

Equalization is therefore not simple subtraction.

It does not remove Momentum one by one until nothing remains.

Each resolution rewrites the relational network.

The rewritten network reveals new Momentum.

Cosmic diffusion may be understood as this process occurring on an extraordinarily broad scale in the early universe.

Because the relational network was so densely coupled, even one change could propagate through many others.

Early cosmic diffusion may therefore have been a far more fundamental network-wide transformation than a simple displacement of matter.

Was the Big Bang a Moment, or a Process?

This also allows us to reconsider whether the Big Bang should be thought of primarily as one instant.

We often speak as if the Big Bang were an event occurring at t=0.

DISTANCE is less concerned with one instantaneous explosion than with a process in which extremely strong relational Momentum was resolved and the network was reconfigured.

If the Big Bang is reread as diffusion, the important thing is therefore not only the first instant but the continuing resolution that followed.

Some Momentum may have been resolved very rapidly.

Some relationships may have persisted much longer.

As new structures formed, new local Momentum may have appeared.

And the global directionality established in the early relational transformation may have remained effective in later cosmic evolution.

The Big Bang can therefore be viewed not merely as a completed event in the distant past, but as the initial phase of a continuing relational transformation that produced the universe we observe today.

Is the Present Universe Still Part of That Diffusion?

When, then, did cosmic diffusion end?

Did it cease once the earliest rapid transformations had passed?

DISTANCE does not need to assume that it did.

Relationships continue to change.

Geometric separations among galaxies change.

Stars form and disappear.

Matter enters new structures and leaves old ones.

Some Momentum are resolved and others become effective.

The relational rearrangement that began in the early universe may therefore, in some form, still be continuing.

This does not mean that the early and present universe have the same intensity of change.

Their relational conditions are radically different.

The distribution of Distances has changed.

The organization of effective Momentum has become far more complex.

But the basic sequence:

Difference → Distance → Momentum → Equalization → resolution → rearrangement

may remain applicable.

This possibility later raises the question of whether cosmic expansion should be regarded merely as an inertial aftereffect of the early Big Bang, or as part of an ongoing global relational process.

DISTANCE Does Not Eliminate the Big Bang

The position of DISTANCE should therefore remain clear.

Rereading the Big Bang as diffusion does not deny the Big Bang.

It does not reject the high-temperature, high-density conditions of the early universe.

Nor does it claim that the universe has always had its present configuration.

What DISTANCE questions is the starting level of explanation.

Must observed cosmic change immediately be grounded in the expansion of an independently existing space as the deepest explanatory layer?

Or can relational transformation be placed conceptually prior to that interpretation?

The question is therefore not:

“Did the Big Bang happen?”

It is:

“At what conceptual level should the cosmic transformation we call the Big Bang be explained?”

DISTANCE begins with relationships.

From Explosion to Diffusion

We can now compare two conceptual pictures.

The first is explosion:

center → outward force → objects flying apart → larger occupied space

The second is DISTANCE's diffusion:

extreme relational density → short effective Distances → strong unresolved Momentum → Equalization → Momentum resolution → increasing relational Distances → network-wide rearrangement → observable separation and emerging structures

The second picture does not require an external space from the outset.

It does not require one center.

It does not require one force pushing everything outward.

It requires relationships, Difference within those relationships, Distance, Momentum, and the reconfiguration produced as Momentum is resolved.

But Expansion Has Not Yet Been Explained

None of this means that the problem of cosmic expansion is already solved.

We have not yet explained in detail the enormous geometric separation observed in the universe.

We have only established another possible starting point from which to interpret it.

Several questions remain.

What exactly is the separation we observe?

How is increasing relational Distance related to increasing geometric distance?

Is the language of expanding space necessary to describe the observations, or is it one interpretation of those observations?

Where does the relational account offered by DISTANCE overlap with the conventional language of expansion, and where does it diverge?

Without addressing these questions, diffusion would remain only a metaphor.

The next section therefore deliberately separates expansion from separation.

The Next Step — Expansion or Separation?

The first purpose of rereading the Big Bang as diffusion is not simply to replace one word with another.

Replacing expansion with diffusion would by itself explain nothing.

The real shift lies in changing the explanatory unit:

from change in space to change in relationships,

from motion of independent objects to relational rearrangement,

from an external first impulse to the resolution of unresolved Momentum.

But this shift requires us to distinguish something carefully.

To what extent is the separation we actually observe the same thing as interpreting that separation as “the expansion of space”?

And at the same time:

To what extent is increasing relational Distance the same thing as increasing geometric separation?

Neither question allows a simple identification.

That distinction is the subject of the next section:

2.3 Expansion or Separation?

3 Expansion or Separation?

When we speak of the expansion of the universe, one of the first images that comes to mind is that distant galaxies are moving farther apart.

The geometric separation between one galaxy and another increases.

More distant galaxies are observed to recede more rapidly.

And modern cosmology describes these observations using the language of an expanding universe.

This description is extraordinarily successful.

DISTANCE, however, asks us to make one distinction.

Are observed separation and the expansion used to explain it statements at the same conceptual level?

The two are deeply related, but they are not identical.

Separation is a change that we observe or infer in the relation between states.

Expansion is a physical and geometric interpretation of that change.

DISTANCE is interested precisely in the gap between these two levels.

What Do We Actually Observe?

When we observe the universe, we do not directly watch space itself stretching.

What we observe are changes in the states and relations of celestial objects.

Spectra change.

Relative distances and recession velocities are inferred.

Geometric relationships among galaxies change.

From these observations, we construct a description of the dynamics of the universe.

Within that description, the idea of space itself expanding provides a powerful and successful framework.

But it remains important to distinguish observation from interpretation.

We encountered the same problem in Chapter 1.

On a computer desktop, an icon genuinely moves.

But the form of that movement is not identical to the internal state changes taking place beneath the interface.

Likewise, increasing geometric separation in the universe may be a genuine and essential feature of observation.

The separate question is whether its deepest explanation must necessarily be that space itself is stretching.

This is the opening that DISTANCE explores.

Separation Is a Relational Change

Consider two galaxies, A and B.

Suppose an observer infers that the geometric distance between them is increasing.

We can call this separation.

But separation is already, in itself, a change in relationship.

The positional relation between A and B changes.

The conditions under which light travels between them change.

Their relative configuration with respect to other celestial objects changes as well.

DISTANCE therefore begins by reading separation as relational change.

The advantage of this approach is that change can be described without immediately treating space as the first cause.

A relationship changed.

The configuration of the wider network changed.

And geometric separation changed as an observable consequence.

Conceptually, we can consider:

relational rearrangement $\rightarrow$ changed observable configuration $\rightarrow$ geometric separation

This does not deny expansion.

It leaves open the possibility that another explanatory layer lies beneath the geometric description.

Geometric Separation and Relational Distance Are Not the Same

At this point, an important caution is required.

If the geometric separation between A and B increases, we cannot automatically conclude that their relational Distance has also increased.

Chapter 1 distinguished the two concepts explicitly.

Geometric distance is separation between positions in space.

Relational Distance expresses how effective a Difference is within a relationship.

Therefore:

geometric distance ≠ relational Distance

Two celestial objects may become geometrically farther apart while some other relationship between them becomes more strongly effective.

Conversely, two objects may move geometrically closer while a particular relational Momentum is being resolved and the corresponding Distance increases.

We must therefore not create a new identity such as:

increase in relational Distance = increase in geometric separation

That would merely replace one simplification with another.

Does That Mean the Two Are Unrelated?

No.

If geometric separation had no stable relation at all to relational conditions, geometric distance would hardly be as successful as it is in physical explanation.

The DISTANCE approach is therefore not to separate the two completely, but to investigate their correspondence.

Relational effectiveness changes.

Momentum is resolved.

The relational network is rearranged.

The geometric configuration changes as a result.

If such transformations occur repeatedly, stable correspondences may emerge between spatial separation and relational state.

What matters is that we do not fix the causal direction in advance.

The familiar explanatory sequence is often:

space expands → objects become more separated → interactions change.

DISTANCE also considers the reverse explanatory possibility:

relationships change → effective Momentum changes → the relational network rearranges → geometric separation emerges or changes

These two descriptions need not be completely incompatible.

But they place the deeper explanatory emphasis in different places.

What Does It Mean to Say That Space Expands?

The phrase space expands is so familiar that we rarely stop to ask what kind of statement it is.

When an ordinary object expands, we can say that its internal structure changes.

If a rubber sheet stretches, the relationships among the constituents of the rubber change.

But when we say that space itself expands, what exactly is changing?

DISTANCE takes this question literally.

If space is treated as an independent substance, then one may describe the size of space itself as changing.

But if space is understood as an observable expression of relational configuration, the statement space expands can be read differently.

It may mean:

the geometric representation of the relational network changes toward a configuration with greater separation.

From this perspective, expansion may be a geometric description of relational rearrangement rather than the underlying mechanism itself.

This is not merely a change in vocabulary.

It reverses the order of explanation.

How Far Does the Balloon Analogy Take Us?

A familiar analogy for cosmic expansion is the surface of an inflating balloon.

Place several dots on the surface and inflate the balloon.

The distances among the dots increase.

From the perspective of any one dot, the others appear to recede.

And no special center exists on the surface itself.

This analogy is very useful for understanding geometric expansion.

DISTANCE, however, asks one additional question.

What is actually changing on the balloon?

The relational state of the rubber changes.

The separations and couplings among its constituents change.

And those relational changes appear geometrically as expansion.

Of course, the universe is not a rubber membrane.

That is not the point.

The useful lesson is narrower:

geometric expansion itself can be represented through relational change.

The balloon analogy may therefore reinforce the DISTANCE question rather than oppose it.

What relational transformation might lie beneath what we observe as cosmic expansion?

The Observer and the Interpretation of Expansion

One of the striking features of cosmic expansion is that it does not identify a simple geometric center from which everything else recedes.

From our location, distant galaxies appear to recede.

An observer in another galaxy would infer a similar pattern.

This is not the ordinary pattern of an explosion from one center.

A conventional explosion allows us to distinguish center from exterior.

A relational interpretation accommodates this feature naturally.

If the relational network as a whole is being rearranged, no single pre-existing center need be assumed.

Each observer measures changes in geometric separation from within the relational configuration in which that observer is embedded.

Many-directional recession can therefore appear from many observational locations.

The important point is:

multi-directional separation does not by itself require one central explosion.

This will become more important when we connect diffusion to Equalization.

An Observer Riding on an Ink Particle

We can make the idea more intuitive by modifying the ink analogy.

Imagine ink diffusing through water.

Now imagine an observer attached to one ink particle.

That observer watches the surrounding ink particles.

As diffusion proceeds, many nearby particles may appear to move away in different directions.

From within the distribution, the pattern might resemble an overall expansion of the observer's surroundings.

Yet the cup itself has not expanded.

What has changed is the relational configuration of the particles within it.

This thought experiment is not intended to reproduce the actual mechanism of the universe.

Its purpose is only to isolate one logical possibility:

many-directional recession can arise from relational rearrangement without requiring the container itself to expand.

That possibility is enough for the present argument.

"Then Is the Universe Just Ink in a Cup?"

No.

The universe does not require an external water-like background.

Its constituents need not correspond to already well-defined ink particles moving through a pre-existing medium.

If the analogy were transferred literally, it would fail immediately.

What DISTANCE borrows from it is only the structure of observation.

An observer located inside a changing relational distribution can observe recession in many directions.

Thus:

an observed recession pattern is not, by itself, a proof of an expanding external container.

Which physical model actually applies to the universe requires further validation.

The Goal Is Not to Eliminate the Word "Expansion"

The position of DISTANCE must remain clear.

The goal is not to forbid the word expansion.

If the language of expansion is useful for organizing observations, making predictions, and carrying out calculations, there is no reason to discard it.

The issue is whether that language should automatically be treated as the final ontological explanation.

We can distinguish the following statements:

The universe is observed as expanding.

and

Space itself is the fundamental entity whose intrinsic expansion causes the observed separation.

The first is close to an observational description.

The second contains a deeper interpretive commitment.

DISTANCE can accept the first while reopening the second.

Separation Is Not Merely the Motion of Independent Objects

Replacing expansion with separation does not solve everything either.

The word separation can still suggest that fully formed independent objects already existed and simply moved farther apart.

But in the early universe, many of the structures we now identify as objects had not yet formed.

Cosmic separation therefore need not mean only that already completed galaxies moved apart.

At a more general level, the process may involve:

relationships differentiating, configurations becoming distinguishable, Islands emerging, and geometric separation becoming measurable.

In this sense, separation itself can be an observable result of relational differentiation.

Separation and Structure Formation Can Occur Together

This perspective also helps clarify an important feature of cosmic evolution.

The universe exhibits large-scale separation while simultaneously forming local structures.

Galaxies may become increasingly separated while stars and planets form within local regions.

If expansion and aggregation are treated only as opposite geometric motions, it is tempting to explain them merely as competing forces.

DISTANCE allows a more relational reading.

A broader relational configuration may be rearranging in one way,

while local Differences become newly effective,

local Distances become short,

and local Momentum emerge.

Thus:

global separation and local binding can coexist.

There is no contradiction.

Different Momentum may become effective at different scales and within different relationships.

Global and Local Are Not Two Independent Worlds

The words global and local can easily suggest two independent mechanisms.

That is not the intended meaning.

Local Momentum does not exist in complete isolation from the global process.

Global relational rearrangement changes the conditions under which local Difference becomes effective.

Local rearrangement, in turn, changes the conditions of the broader network.

The two levels are therefore hierarchically and dynamically connected.

Chapter 3 will develop this relationship more carefully.

For Chapter 2, it is enough to retain one point:

overall diffusion does not eliminate local structure formation.

Could the Expansion Rate Be a Scale of Relational Change?

Even what we call the expansion rate can be reconsidered relationally.

Conventionally, the expansion rate describes how geometric separation changes.

DISTANCE cannot simply identify this with the rate of change of relational Distance.

But if broader relational rearrangement is reflected in geometric configuration, then the expansion rate may function as an observable scale of underlying relational transformation.

This reconnects with the argument of Chapter 1.

Time may serve as a scale through which the intensity of change becomes comparable.

Space may serve as a geometric expression of relational complexity.

The expansion rate may therefore be interpreted as one way in which relational reconfiguration becomes measurable within a spacetime description.

The claim here remains one of correspondence, not identity.

Redshift Must Be Treated the Same Way

Redshift is one of the central observational foundations of cosmic expansion.

DISTANCE has no reason to ignore such observations.

But redshift itself and the interpretation of redshift as spatial expansion should not be treated as exactly the same statement.

Redshift is an observation.

Connecting it to cosmic expansion is part of a physical interpretation.

DISTANCE does not simply reject that interpretation.

It asks:

How might relational rearrangement correspond to the same observational pattern?

Answering that question rigorously would require quantitative validation.

At this stage, the argument must therefore remain limited enough not to claim more than the framework can yet establish.

A Better Explanation Must Preserve Observation

This point also clarifies how DISTANCE relates to science.

A new interpretation cannot simply discard successful observations.

Those observations must be preserved.

A deeper explanation would have to account for the same empirical regularities while placing them within another explanatory structure.

The goal of DISTANCE is therefore not to build a competing cosmological story by ignoring conventional cosmology.

Its more fundamental question is:

What relational mechanism might underlie the current observational description?

And eventually:

Can that mechanism reproduce the observations with equal or greater explanatory coherence?

Expansion and Diffusion May Belong to Different Levels

We can now stop treating expansion and diffusion as two labels competing for the same level of description.

Expansion may be the language in which the phenomenon is described within a geometric framework.

Diffusion may be the language with which DISTANCE attempts to describe the underlying relational transformation.

In that case, the two need not be mutually exclusive.

A layered relation such as:

relational diffusion → observable geometric expansion

becomes conceptually possible.

This remains a hypothesis.

But the important point is that expansion and diffusion need not be treated as rival words for exactly the same thing.

One belongs more naturally to observational geometry.

The other is intended as an account of relational mechanism.

Is Separation More Fundamental Than Expansion?

Another question then appears.

If we use separation instead of expansion, have we reached a more fundamental description?

Not necessarily.

Separation can still remain a geometric observation.

Within DISTANCE, the deepest level considered here lies beneath separation as well:

Difference → Distance → Momentum → Equalization → relational rearrangement

Separation may emerge as one observable result of that process.

The hierarchy can therefore be expressed approximately as:

relational change → configuration change → possible geometric separation → expansion description

Keeping this hierarchy prevents separation itself from becoming another unexplained fundamental entity.

Expanding Space May Be a Successful Interface

The Fake Desktop argument from Chapter 1 can be brought back here.

The space of a computer desktop is not identical to the structure of the hardware beneath it, yet it is extremely effective.

Likewise, the description of expanding space might be extraordinarily accurate as an interface even if the underlying relational process is described differently.

Through that description, observations can be organized,

predictions made,

and mathematical calculations performed.

Its success matters.

But success does not by itself prove that the observational form is identical to the underlying ontology.

That distinction was established in Chapter 1.

Chapter 2 is applying it to cosmology.

Reading Cosmic Separation in Relational Terms

We can now give a provisional DISTANCE reading of cosmic separation.

The early relational network contained very short effective Distances and strong unresolved Momentum.

Those Momentum began to be resolved.

Relational Distances changed.

The network was rearranged.

Configurations differentiated.

And geometric separation could emerge as an observable result.

We may therefore consider the hypothesis:

cosmic separation is an observable geometric expression of a deeper relational rearrangement.

But this must not be read as an absolute identity.

Not every separation necessarily arises from the same mechanism.

And not every increase in relational Distance must appear as geometric separation.

The present argument concerns a possible correspondence within the specific context of early cosmic diffusion.

What Is Actually Expanding?

The question can now be reformulated.

What is actually expanding?

Space?

Relationships?

The geometric representation of relationships?

Or does the distinction itself depend on the scale at which we observe the process?

DISTANCE v1.1 does not attempt to settle this completely here.

It makes a more limited commitment.

Observed recession and geometric separation should not automatically be frozen as the final explanatory layer.

The relational process beneath them remains open to investigation.

The Next Step — Diffusion as a Path of Equalization

Section 2.2 reread the Big Bang as diffusion rather than simple explosion.

Section 2.3 has now separated expansion from separation.

The next question follows.

Why does one relational rearrangement become more effective than another?

When countless Momentum coexist, which resolution paths become prominent?

And how should the term preferred path be used without once again reversing the direction of Distance?

This is the central problem of the next section:

2.4 Diffusion as a Path of Equalization

There we will apply directly to cosmic diffusion the principle that shorter Distance is not the destination of Equalization, but a relational condition associated with stronger effective Momentum.

4 Diffusion as a Path of Equalization

In the previous sections, we separated two things.

First, the vast separation we observe after the Big Bang need not be identified immediately with the expansion of space itself.

Second, an increase in relational Distance is not the same thing as an increase in geometric separation.

One question now remains.

If the early universe contained countless Differences and Momentum at the same time, why did change not simply unfold randomly in every possible direction?

Why did some relationships participate in change earlier or more strongly than others?

And when DISTANCE rereads the Big Bang as diffusion, by what principle did that diffusion proceed?

This is ultimately a question about the paths of Equalization.

Not Every Relationship Changes to the Same Degree

Consider the early universe as a relational network.

There is a Difference between A and B.

There is another Difference between A and C.

There are still others between B and C, C and D, D and E, and countless additional relations.

There is no reason to assume that all of these relationships are equally effective.

Some Differences may be strongly effective within a particular relationship.

Others may exist but produce little effect under the present relational conditions.

The fact that cosmic change began therefore does not imply that every relationship changed simultaneously with the same intensity or at the same rate.

Some relationships were more strongly effective.

Others were less strongly effective.

And, as defined in Chapter 1, DISTANCE describes this distinction by saying:

a more strongly effective relationship has a shorter Distance.

A shorter Distance does not simply mean that the two states are spatially closer.

It means that the relationship is more capable of participating strongly in the change currently taking place.

Diffusion must therefore be understood not as one uniform process distributed evenly across the entire relational network, but as a process in which countless resolution paths with different relational effectiveness coexist, compete, and continually adjust one another.

A Preferred Path Is Not a Destination

This is where the word preferred becomes useful.

If one relationship is more strongly effective than another, its directionality may become more prominent in actual change.

We may call this a preferred path.

But the word must be used carefully.

Preferred does not mean that the universe knows which path is better and chooses it.

It does not imply that the universe calculates an optimal route toward a goal.

More precisely, it means:

within the present relational configuration, a path carrying stronger effective Momentum becomes more prominent in actual change.

A preferred path therefore expresses relational priority, not teleological selection.

This distinction is essential.

Shorter Distance Is Not the Result of Equalization

This is also where the direction of DISTANCE is most easily reversed.

When expressions such as preferred Equalization and shorter Distance appear together, it is easy to read them as meaning:

Equalization proceeds toward shorter Distance.

Or:

As Equalization progresses, Distance decreases.

But Chapter 1 established the opposite direction.

A shorter Distance is not the result of Equalization.

It is a starting condition indicating that the relationship is presently more strongly effective.

Thus:

shorter Distance → stronger effective Momentum → more prominent Equalization path

Once that Momentum begins to be resolved, however, the direction changes:

Momentum resolution → weaker relational effectiveness → increasing Distance

The concepts of preference and resolution must therefore not be read as pointing in the same direction.

Preference tells us which relationship is presently more effective.

Resolution tells us how that relationship changes as its Momentum is resolved.

The Risk of the Phrase "Minimum Distance"

The same caution applies to the expression minimum Distance.

Suppose one relationship has a shorter Distance than the alternatives currently available.

That relationship carries stronger effective Momentum.

It may therefore participate more prominently in Equalization.

In this limited sense, we might describe it as a minimum-Distance path.

But this must not mean:

the universe aims at a minimum-Distance state.

The more accurate meaning is:

among currently available relationships, the one with shorter Distance carries stronger effective Momentum and may therefore emerge as a more prominent path of Equalization.

Minimum Distance is therefore not a destination.

It is the state of a relationship that is presently most strongly effective.

If this distinction is lost, the entire direction of Big Bang diffusion is reversed.

The Starting Condition and the Direction of Progress Must Be Separated

The argument can be simplified into two distinct questions.

The first concerns the starting condition.

Which relationship is more strongly effective?

If one relationship has a shorter Distance, its effective Momentum is stronger.

The second concerns the direction of progress.

When that Momentum is resolved, relational effectiveness weakens and Distance increases.

Thus:

shorter Distance → stronger Momentum

describes the present relational condition.

Whereas:

Momentum resolution → increasing Distance

describes the direction in which that relationship changes through resolution.

These statements are not contradictory.

The first answers:

“Which path is more strongly open to change now?”

The second answers:

“What happens to that relationship as change proceeds along that path?”

Diffusion Is Not One Straight Path

If the early universe contained only one relationship, the story would be simple.

The A-B relationship is the most strongly effective.

Its Momentum is resolved.

Distance increases.

And the process ends.

But the actual universe contains countless relationships.

When A-B changes, A-C changes.

B-D changes.

C-E may become newly important.

A relation that was previously weak may suddenly become more effective.

Its Distance may become relatively shorter, and a new Momentum may emerge.

Diffusion therefore does not proceed along one preferred path fixed from beginning to end.

The preferred path itself can change as the relational network changes.

This point is crucial.

The path of Equalization is not a line drawn in advance.

As relationships change, the Momentum that becomes effective also changes, and the next path of resolution is reconstituted.

When Relationships Change, Their Priorities Change

Suppose A-B is initially the strongest relationship.

Part of its Momentum is resolved.

A and B change.

As a result, A-C may become more strongly effective.

Its Distance may now become relatively shorter.

The A-C relationship may therefore emerge as a new preferred path.

Diffusion may consequently proceed through a sequence such as:

one preferred relationship → partial resolution → network rearrangement → another preferred relationship

But in the early universe, these transformations almost certainly should not be imagined as occurring neatly one after another.

Countless relations change simultaneously.

Countless Momentum weaken and strengthen at the same time.

The better picture is one of continuously recomputed relational priority.

Equalization Is Not a Shortest-Path Search

The phrase preferred path can easily suggest another misunderstanding.

The universe may begin to sound like an optimization algorithm searching all available routes and selecting the most efficient one.

DISTANCE assumes no such calculating agent.

If one Momentum is more strongly effective within a given relational configuration, that direction simply becomes more prominent in actual change.

Nothing more is required.

Equalization is therefore not:

a process that evaluates all possible paths and deliberately selects the best one.

It is:

a process in which differences in effective Momentum make some relational paths more prominent than others under the present relational conditions.

This distinction keeps the argument non-teleological.

Diffusion Does Not Need One Direction

An explosion suggests a center and an outward direction.

Diffusion as a path of Equalization does not require one absolute direction.

One relationship may carry Momentum in one direction.

Another may carry Momentum in another.

These Momentum coexist.

Diffusion can therefore proceed through many directions at once.

This does not mean that every local direction is random.

Each direction arises from the Difference, Distance, and broader relational configuration of that relation.

The overall process can therefore be understood as:

many local resolution paths within one broader relational transformation

Cosmic diffusion is, in this sense, much more complex than a simple outward explosion.

Where Does Global Directionality Come From?

If countless local paths point in different directions, how can there still be an overall direction to diffusion?

Chapter 3 will later develop this question through the distinction between global and local Momentum.

For now, only a minimal observation is needed.

The early universe shared one broad condition:

extreme relational concentration.

Many relationships had short Distances.

Many Momentum were strongly effective.

And that extreme concentration did not remain unchanged.

In this broader sense, there was a common direction of transformation: the resolution of extreme relational concentration.

Local Momentum need not point in the same geometric direction for this broader tendency to exist.

A global directionality does not mean one gigantic vector shared identically by every local relation.

It means a common direction of resolution at the level of the wider relational network.

Can We Say That Closer Relationships Resolve First?

We can now use one expression cautiously:

"Closer relationships resolve first."

This can be useful, but only if interpreted precisely.

What it really means is:

relationally shorter-Distance relationships tend to carry stronger effective Momentum and therefore become more prominent in ongoing Equalization.

The word first does not imply that relations are processed one by one in a temporal queue.

In the early universe, countless relationships changed simultaneously.

Preferred therefore refers to relative effectiveness, not strict temporal ordering.

Why Does Distance Become Longer Through Resolution?

Let us verify the direction one more time.

Suppose the A-B relationship is strongly effective.

Its Distance is short.

As its Momentum is resolved, the Difference no longer acts through that relationship as strongly as before.

Its relational effectiveness weakens.

Distance therefore becomes longer.

In this sense:

resolution = relational weakening

But relational weakening does not automatically mean that A and B become geometrically farther apart.

Geometric approach may occur while a relational Momentum is being resolved.

Thus:

Distance increase is relational, not automatically geometric.

The distinction remains essential in cosmic diffusion.

Not Every Distance Increases

While one relationship is resolved and its Distance increases, another relationship may become more strongly effective.

A-B may become more distant while A-C becomes closer.

B-D may become newly important.

The Equalization of the universe therefore cannot be described as:

every Distance everywhere continually increases.

A more accurate statement is:

each resolved Momentum tends to increase the Distance of the relationship through which it was effective, while the resulting rearrangement may shorten other Distances and create new effective Momentum.

This prevents Equalization from being misread as a process in which all relations simply weaken monotonically.

Then Why Do Structures Form?

This naturally leads to the next problem.

If cosmic diffusion resolves Momentum and increases the Distance of the corresponding relationships, why do structures form at all?

Why do stars form?

Why do galaxies form?

Why does matter bind into relatively persistent configurations?

The question does not contradict the argument developed so far.

When one relationship is resolved, another may become shorter.

As the global configuration changes, local Differences may become newly effective.

New local Momentum may emerge.

New binding relationships may become possible.

Equalization therefore does not necessarily destroy all structure.

It may instead open new paths through which structures can form.

But this is the central problem of Section 2.5, so we do not develop it fully here.

Preferred Paths and Structure Formation May Be Connected

One connecting idea can nevertheless be left in place.

Suppose relational rearrangement causes a certain local relationship to acquire a relatively shorter Distance.

That relationship now carries stronger effective Momentum.

It may therefore become more prominent in subsequent Equalization.

A set of constituents may then begin to form a persistent configuration.

Structure formation may therefore not require a mechanism entirely outside Equalization.

A local binding path may emerge from within the relational rearrangement produced by Equalization itself.

The next section will develop this possibility directly.

Does Diffusion Mean Optimization?

As we use terms such as rearrangement, preferred, and effective, the language of optimization may become tempting.

Relationships become more or less effective.

The configuration continually changes.

Priorities are recomputed.

But optimization can also sound teleological.

If the word reoptimization is used later, it should therefore be understood in a limited sense:

relational conditions change, causing the effective priority of relationships to be continuously recomputed.

It does not automatically imply that the universe is minimizing or maximizing some fixed objective function.

In particular, it must never be read as an optimization that minimizes Distance itself.

Avoid Saying "The Universe Searches for Shorter Distance"

For the same reason, expressions such as the following should be avoided:

the universe searches for shorter Distance

or:

Equalization moves toward minimum Distance.

Such phrases easily conflict with the definitions established in Chapter 1.

The more precise statement is:

shorter-Distance relationships become more effective paths of change because they carry stronger Momentum.

And as Momentum is resolved along such a path:

that relationship moves toward longer Distance.

These two statements must remain paired.

Diffusion Continually Redirects Itself

The picture of cosmic diffusion can now be made more precise.

The early universe contained countless strong Momentum.

Some relationships were more strongly effective than others.

Those relationships became more prominent in change.

Momentum was resolved.

Distance increased.

The relational network was rearranged.

Other relationships acquired relatively shorter Distances.

New Momentum became effective.

And change continued.

Cosmic diffusion can therefore be described as:

a continuously redirecting process of Momentum resolution.

A single outward flow is not enough to describe it.

Equalization Does Not Impose One Path on the Entire Universe

From this perspective, Equalization is not a global force commanding every constituent to move in the same way.

Each relationship has its own Difference and relational conditions.

Each Momentum has its own directionality.

Yet these relationships are not independent.

A change in one relation alters the conditions of others.

Equalization is therefore better understood as:

many relational paths coupled through one changing network.

This allows diversity of local directions and coherence of a broader process to coexist.

Diffusion Does Not Require a Predetermined Final State

If preferred paths continually change, does Equalization still require one fixed final state?

There is no need to assume so at this stage.

When one Difference is resolved, relational conditions change.

New Momentum may become effective.

The configuration itself changes.

There is therefore no basis for assuming that a final state is specified in advance and that every transformation simply moves toward it.

Equalization is:

the continuing resolution of Momentum that is effective within the present relational configuration.

The configuration changes through that resolution.

A destination need not be predefined.

Diffusion and the Present Universe

This view can also be extended to the present universe.

Some relationships remain strong.

Others are weak.

Some Momentum are being resolved.

Others become newly effective.

Cosmic diffusion may therefore not be a special process restricted only to one instant in the early universe.

The global relational rearrangement that began as extreme relational concentration was resolved may continue in other forms and at different intensities.

This connects with the view of the Big Bang not merely as a completed explosion, but as the early phase of an ongoing process of Equalization.

Now the Problem of Structure Appears

Yet one major feature of the universe remains unexplained.

The universe does not contain only separation.

It contains structure.

There are stars.

Galaxies.

Planets.

And countless objects that remain bound for relatively long periods.

If Equalization resolves Momentum and increases the Distance of the corresponding relationships, how can such bound structures exist?

Are structures failures of Equalization?

Or can they themselves emerge as particular forms of relational rearrangement within continuing Equalization?

This is the central question of the next section:

2.5 Structure Within Continuing Equalization

There we will examine why cosmic diffusion should not be reduced to simple dispersion or increasing disorder, and how new Islands and Structures may emerge within a continuously rearranging relational network.

5 Structure Within Continuing Equalization

If we carry the argument of the preceding sections forward, we encounter what initially appears to be a paradox.

The early universe was densely packed with relationships characterized by extremely short relational Distances.

Those relationships carried strong Momentum.

Momentum is resolved through Equalization.

And when the Momentum associated with a particular relationship is resolved, the effectiveness of that relationship weakens and its Distance increases.

We have interpreted this vast transformation of relationships as cosmic diffusion.

Yet the universe we actually observe is not a state in which everything has simply dispersed.

There are stars.

There are planets.

There are galaxies.

Countless constituents remain bound together and preserve recognizable configurations for extended periods.

Some structures persist for billions of years.

This raises an obvious question.

If change proceeds through the resolution of Momentum, why does structure exist at all?

Is structure something that survives by resisting Equalization?

Is it an exception produced because cosmic diffusion has remained incomplete?

Or is what we call structure itself something that forms within the process of Equalization?

DISTANCE focuses on the third possibility.

Structure may not be the opposite of Equalization, but a temporary relational configuration that emerges within continuing Equalization.

The Habit of Opposing Dispersion and Structure

We commonly think of diffusion and structure as processes moving in opposite directions.

On one side, things gather.

On the other, they disperse.

Gathering creates structure.

Dispersion destroys it.

If this intuition is applied directly to the universe, cosmic diffusion after the Big Bang and the formation of stars and galaxies appear to be opposing processes.

The universe spreads on the large scale,

while matter gathers locally.

We are therefore tempted to imagine expansion operating in one direction while gravity works against it to create structure.

At the level of conventional physical description, this distinction is extremely useful.

DISTANCE, however, asks a question one level below it.

Are these two phenomena really opposite processes at the relational level as well?

As we saw in the previous section, global diffusion does not mean that every relational Distance increases simultaneously.

When one Momentum is resolved, the relational network changes.

When the network changes, other Differences become effective.

If a Difference becomes strongly effective in a newly configured relationship, a new short Distance appears.

And new Momentum becomes effective.

The resolution of one relationship and the formation of another can therefore occur within the same process.

One Resolution Changes the Wider Network

Consider a simple relational network.

There is strong Momentum in the A-B relationship.

That Momentum is resolved.

The Distance of the A-B relationship increases.

If we stop there, it appears that one relationship has simply weakened.

But A and B themselves have changed through that process.

The A-C relationship therefore need not remain as it was.

The B-D relationship may also change.

Even the conditions of the C-D relationship may be affected indirectly.

Thus:

Momentum resolution does not merely remove a relation. It changes the conditions of other relations.

This point is fundamental.

If Equalization is imagined as a process that simply eliminates existing Momentum one after another, the final universe begins to look like a state in which every relationship has become weak.

But in an actual relational network, the resolution of one Momentum changes the conditions under which other Momentum become effective.

A sequence such as the following can therefore continue:

resolution → rearrangement → newly effective Difference → new Momentum

Equalization is not merely elimination.

It is also rearrangement.

A Momentum May Be Resolved, but Momentum Does Not Disappear

This distinction is particularly important within DISTANCE.

A particular Momentum can be resolved.

That does not mean that Momentum as such disappears from the universe.

As long as relational networks remain and Differences continue to be effective, other Momentum continue to emerge.

As established in Chapter 1:

The existence of Momentum is never the question. Momentum is always present.

Equalization must therefore not be understood as a process that gradually removes Momentum from the universe until a state of perfect stillness is reached.

When one Momentum is resolved, the configuration changes.

Within that changed configuration, other Momentum become effective.

The observable universe is therefore the present expression of which Momentum are effective at a given relational state.

From this perspective, structure is not a place where Momentum has disappeared.

It is a place where multiple Momentum coexist in a configuration capable of persisting for some interval.

Structure Is Not a State in Which Change Has Stopped

When we look at a structure, we tend to imagine something static.

A stone appears to remain still.

A planet retains a recognizable form.

A star can persist as one celestial object for an enormous period.

Structure therefore appears to be a state in which change has somehow stopped.

But the picture changes when the observational scale changes.

Countless transformations continue inside a structure.

Its constituents remain in relation.

It exchanges energy with its surroundings.

It interacts with other structures.

Internal Momentum continue to exist.

A structure therefore cannot be defined in DISTANCE as:

a state without Momentum.

It is more appropriately understood as:

a relatively persistent configuration of coexisting Momentum.

Structure is not the absence of change.

It is a state in which change proceeds while maintaining a recognizable configuration for some period.

Persistence Is Not Stasis

A whirlpool offers a useful intuition.

A whirlpool forms in flowing water.

The water continues to move.

The molecules constituting the whirlpool continually change.

Yet for some period, we can recognize the whirlpool as one persistent form.

Its persistence does not mean that motion has stopped.

In fact, its recognizable form can persist precisely because motion continues.

This is not to claim that stars or planets are physically identical to whirlpools.

The important distinction is between persistence and stasis.

Something can persist without the Momentum within it disappearing.

Structure in DISTANCE possesses persistence in precisely this sense.

An Island of Smaller and More Complex Momentum

A comparison between the extreme conditions of the early universe and the structures of the present universe reveals an important difference.

In the early universe, many relationships had extremely short Distances, and the corresponding Momentum were extraordinarily strongly effective.

As cosmic diffusion proceeded, much of this Momentum was resolved.

The Distances of those relationships increased.

Yet through that process, the relational network was rearranged into increasingly diverse configurations.

New local Differences became effective.

New relationships formed.

The directions of effective Momentum diversified.

As a result, some regions could develop configurations in which no single extreme Momentum dominated as before, but multiple smaller and more complex Momentum mutually constrained and reorganized one another.

When such a configuration persists for some interval, an observer may recognize it as one structure.

In the language of DISTANCE, it can be called an Island.

Thus:

A persistent Island is not a region where Momentum has disappeared, but a temporarily sustained configuration in which Momentum has become smaller, redistributed, and more complex.

Here, smaller should not be read as a universal quantitative law claiming that every individual Momentum in every present-day structure must be numerically smaller than every Momentum in the early universe.

The point is that the extreme concentration of early Momentum is progressively resolved, while its directionality becomes redistributed through a more differentiated and complex relational network.

An Island Is Not a Place That Escaped Equalization

The term Island can introduce another misunderstanding.

An island appears separated from the surrounding sea.

It is therefore easy to imagine an Island in DISTANCE as an independent region somehow detached from global Equalization.

That is not what the term means.

No physical structure is completely independent of the broader relational network.

A star has relationships with other stars.

A planet has relationships with its star.

A galaxy has relationships with other galaxies.

And the constituents within every structure continue to form relationships among themselves.

The boundary of an Island therefore does not imply absolute isolation.

It marks a range within which certain relationships are relatively more strongly coupled than others, producing a configuration that can be observed as one recognizable unit.

An Island has not escaped Equalization.

It is a local configuration formed within the broader process of Equalization.

Global Diffusion and Local Binding

We can now return to the problem left open in the previous section.

Diffusion proceeds across the broader universe.

At the same time, binding occurs within some local regions.

These processes need not contradict one another because different Momentum can become effective at different relational scales.

As Momentum in the broader relational network is resolved, relational conditions change.

A particular local Difference may then become more strongly effective.

Its relational Distance becomes relatively shorter.

New local Momentum becomes effective.

And through those Momentum, constituents may form a new persistent configuration.

Conceptually:

global Momentum resolution → relational rearrangement → newly effective local Difference → shorter local Distance → stronger local Momentum → persistent local configuration

Local structure therefore need not oppose global Equalization.

Global rearrangement can itself create the relational conditions under which local structure becomes possible.

But Local Structure Is Not Permanent

Once an Island forms, however, it does not become eternal.

It too is a relational network.

Its internal Differences change.

Its relationships with the outside change.

New Momentum may become effective.

Existing binding Momentum may be resolved.

The persistence of structure is therefore always conditional.

A configuration may persist for a very long time without becoming an absolute final state.

Stars form and disappear.

Planetary systems change.

Galaxies interact and are reconfigured.

Every Island exists within a broader relational rearrangement.

Thus:

structure is persistent, not permanent.

This distinction will become even more important when we later turn to life.

A living system, too, is not a static object, but an Island that continuously changes relationships in order to maintain a recognizable configuration.

Structure Formation Need Not Be a Reversal

We often imagine that when something begins to gather within a dispersing system, the original process has somehow reversed.

Diffusion proceeds, and then gravity reverses part of it by pulling matter back together.

A local event appears to move against the broader direction.

At the relational level, however, DISTANCE does not require this interpretation.

As relationships are rearranged through broader Equalization, new local Momentum may emerge.

If those Momentum appear as local binding, that binding is also part of the ongoing relational process.

Structure formation can therefore be read not as:

diffusion → reversal → structure

but as:

diffusion → rearrangement → new Momentum → local binding → structure

The distinction may appear small, but it is conceptually important.

It prevents structure from becoming an exception to the broader tendency of the universe.

Setting Aside the Language of Order and Disorder

At this point, entropy naturally comes to mind.

Diffusion is often associated with increasing disorder,

while structure formation appears to increase order.

This immediately raises a familiar question:

If the universe as a whole tends toward greater disorder, why do complex structures such as stars and life emerge?

Conventional thermodynamics has sophisticated ways of addressing this problem.

DISTANCE does not deny them.

But this book is testing a different language.

The central question here is not whether the universe has become more ordered or more disordered.

It is:

Which relationships have been resolved? Which relationships have become newly effective? How has the overall configuration been rearranged?

Order and disorder may describe how an observer evaluates a configuration at a particular scale.

DISTANCE does not need to make them the fundamental principle of change.

Rearrangement, Not a March Toward Disorder

The transformation from the early universe to the present universe is therefore difficult to reduce to:

ordered → disordered

or, conversely:

simple → complex.

The extreme relational concentration of the early universe was resolved.

Many relationships weakened.

At the same time, other relationships emerged.

The distribution of Momentum changed.

A variety of persistent configurations appeared.

The central concept is rearrangement.

Some configurations persist only briefly.

Others remain for enormous periods.

Some combine with other structures.

Others dissolve.

What we call the history of the universe can therefore also be viewed as the continuous association, separation, and reconfiguration of such relational configurations.

The Limited Meaning of Reoptimization

This continuing rearrangement may sometimes be described as reoptimization.

But, as emphasized in the previous section, the term must be used carefully.

Within DISTANCE, reoptimization does not mean that the universe knows some final objective function and searches for the optimal state.

In particular, it does not mean that the universe seeks:

minimum Distance.

Its meaning here is much more limited.

As relationships change, the distribution of effective Momentum changes.

Paths that were previously prominent weaken.

Other paths become stronger.

Relationships change again.

And relational priorities are again reconfigured.

Thus:

reoptimization = continuing reconfiguration of effective relational paths

is the safer interpretation.

Is Structure Merely a By-product of Equalization?

Does this mean that structure is merely an accidental by-product?

That conclusion would also go too far.

DISTANCE does not claim that the universe performs Equalization in order to create stars, planets, or life.

That would introduce teleology.

But neither is it necessary to say that structures appear without relational cause.

When rearrangement makes certain local Momentum strongly effective, persistent configurations can form through those Momentum.

Structure is therefore a possible outcome of Equalization.

It is neither its purpose nor its failure.

It is one of the configurations permitted by the changing relational network.

A Single Structure Contains Countless Momentum

Consider a star.

We call it one object.

But from the perspective of DISTANCE, countless relationships exist within it.

Its constituents possess different Differences.

Many Momentum are simultaneously effective.

Some Momentum contribute to maintaining binding relationships.

Others may contribute to transformations within the structure.

The star also participates in relationships outside itself.

A star therefore cannot be reduced to a single Momentum.

A structure is a configuration of multiple coexisting Momentum.

This connects directly with the principle established in Chapter 1:

The universe is never composed of independent objects acted upon by independent Momentum. It is composed of countless coexisting Momentum whose continuously changing balance determines the observable directionality of every physical system.

Stars,

planets,

and galaxies are no exceptions.

Balance Is Not Static Equilibrium

The word balance also requires care.

To say that multiple Momentum form a balance may suggest perfect equilibrium in which everything cancels to zero.

That is not necessarily what DISTANCE means.

Some Momentum may be stronger.

Others may be weaker.

Their relative proportions may continually change.

Yet their relationships can remain within a range that allows the overall configuration to persist.

The persistence of structure therefore arises not from zero Momentum, but from a dynamically maintained relational configuration.

The distinction is essential.

The Observer Sees a Structure

The scale of observation also matters when we distinguish one relational network as a structure.

From a great distance, a star may appear as a single point.

At another scale, it is an enormous system of internal dynamics.

An atom may appear as one object at one scale and as a complex relational configuration at another.

Structure therefore need not mean an absolute unit whose boundary was permanently drawn into the universe from the beginning.

An object becomes observable when a relatively persistent relational configuration can be distinguished as one unit at a given scale.

This is another reason DISTANCE uses the concept of an Island.

An Island is not an absolutely independent entity.

It is a configuration with sufficient relational persistence to become observable as a distinguishable unit.

Boundaries Emerge from Differences in Relationship

If a structure is an Island, it must in some sense possess a boundary.

But that boundary need not always be a rigid geometric surface.

Some relationships may be relatively strongly effective within a configuration,

while relationships across its surroundings operate differently.

Through this contrast, one configuration becomes distinguishable from another.

A boundary can therefore be provisionally understood as:

a transition in relational effectiveness.

The specific physical forms of boundaries will be examined more carefully in Chapter 4.

For now, it is enough to recognize that a structure need not be a perfectly closed container.

Does a Universe with More Structures Become Simpler or More Complex?

The early universe was characterized by an extreme concentration of strong relationships.

The present universe contains countless distinguishable Islands.

Which state is more complex?

The question has no simple answer.

The early universe may have possessed extreme relational density, while its relationships were organized in ways very different from those of the present universe.

The present universe has resolved much of that early Momentum, yet it contains immense varieties of local relationships and hierarchies.

Equalization therefore cannot simply be described as a process that reduces complexity.

As some strong Momentum are resolved, the configurational complexity of the relational network may increase.

This becomes an important bridge to the later discussion of life and society.

The resolution of Momentum and the increase of complexity need not be opposites.

The Universe Is Not a Completed Structure

There is therefore little reason to regard the present universe as a finished final configuration.

The stars, galaxies, and planets that exist today are not eternal end states.

They are Islands maintained under present relational conditions.

Some Islands collapse.

Some combine with other Islands.

Some processes generate newly effective Momentum.

New structures emerge.

The present universe is therefore not what remains after Equalization has finished.

The present universe is the configuration of a moment within continuing Equalization.

This is an important transition in the argument of Chapter 2.

Why Do Some Islands Persist?

A more difficult question now appears.

Why do some configurations disappear almost immediately while others persist for extremely long periods?

Why do some Islands appear stable?

Why do others collapse?

When, and under what relational conditions, do new Momentum become effective?

These questions are important, but they should not all be answered here.

The concrete physical mechanisms through which structures persist and remain bound belong partly to the discussion of Chapter 4.

Much later, we will also examine how living systems maintain their configurations in ways far more complex than ordinary physical Islands.

For Chapter 2, the conclusion can remain more limited:

The existence of structure is not itself a contradiction of Equalization.

The Relationship Between Equalization and Structure

The argument can now be compressed into a single sequence.

The early universe contained strong unresolved Momentum.

As those Momentum were resolved, the Distances of the corresponding relationships increased.

But each resolution rearranged the relational network.

Within that rearranged network, new Differences became effective.

New shorter Distances could appear.

New Momentum emerged.

And some configurations of Momentum persisted for significant intervals.

Thus:

Momentum resolution → increasing Distance in the resolved relationship → relational rearrangement → newly effective Differences → new Momentum → persistent configurations

What we call structure appears within this continuing process.

Therefore:

Structure is not outside Equalization. Structure is one of the configurations that can emerge within continuing Equalization.

Diffusion Continues

The formation of structure does not mean that cosmic diffusion has ended.

Relationships continue to change inside structures.

Relationships among structures continue to change.

New Momentum become effective.

Existing Momentum are resolved.

Diffusion and structure are therefore not two sequential stages in cosmic history.

Diffusion does not first end so that structure can begin.

The two can coexist.

More precisely, structure itself is a temporarily persistent configuration within continuing diffusion and relational rearrangement.

Not a Dispersed Universe, but a Rearranged Universe

It is therefore insufficient to describe the post-Big-Bang universe simply as a universe that became more dispersed.

The extreme relational concentration of the early universe was resolved.

But the result was not the disappearance of relationships.

Relationships were reconfigured.

Momentum was redistributed.

New Islands emerged.

And new networks of relationships formed among those Islands.

The present universe is therefore not merely a more dispersed version of the early universe.

From the perspective of DISTANCE, it is better described as:

a continuously rearranged universe.

The Next Question — What, Then, Is an Object?

The final question of Chapter 2 now follows naturally.

Why do we call one star a star?

Why do we call one planet a planet?

Why do we distinguish a stone from its surroundings and call it one object?

Within DISTANCE, none of these is a completely independent substance.

Nor is any of them a static lump from which Momentum has disappeared.

What, then, is the thing we call an object?

The argument developed so far suggests one possibility:

What we call an object may be a configuration in which countless Momentum form a recognizable pattern that persists long enough to become observable as one unit.

An object, then, is not the opposite of Momentum.

The object itself may be an observable Momentum structure.

And these objects, too, can be interpreted within DISTANCE as Islands.

This leads to the final section of Chapter 2:

2.6 Objects as Observable Momentum Structures

6 Objects as Observable Momentum Structures

In the previous section, we examined why structure need not be understood as the opposite of Equalization.

As strong Momentum in the early universe were resolved, the Distances of some relationships increased. Yet this resolution did not simply eliminate relationships. It rearranged the relational network as a whole, making new Differences effective and generating new Distances and Momentum.

As a result, some relational configurations can persist longer than others.

We recognize such relatively persistent configurations as structures.

Within DISTANCE, these can be called Islands.

One final question therefore remains for Chapter 2.

What is the thing we call an “object”?

We normally begin with objects.

There is a star.

There is a planet.

There is a stone.

And we imagine that these objects move, collide, and influence one another.

But the argument developed so far allows us to reverse that explanatory order.

First, there are relationships.

There is Difference.

There is Distance.

There is Momentum.

As Momentum is resolved and relationships are rearranged, certain configurations persist for some interval.

And we may distinguish those persistent configurations as objects.

From this perspective, the object is not the starting point of explanation.

It is an observable result of a relational process.

Is an Object an Independently Completed Entity?

In everyday experience, objects appear obvious.

A cup on a desk is a cup.

A stone is a stone.

A star is a star.

The inside and outside of its boundary seem distinguishable.

We therefore naturally assume that objects exist first and relationships arise afterward.

There is an object A.

There is an object B.

Then A and B enter into a relationship.

But once Difference was defined in Chapter 1 as a relational concept, this order had already begun to weaken.

Difference is not completed within one isolated state.

Distance is relational effectiveness.

Momentum is relational directionality.

Why, then, must what we call an object be placed prior to that relational network as something completely independent of it?

DISTANCE opens another possibility.

An object may be not an independent entity that subsequently enters into relationships, but an observable result of many relationships forming a sufficiently persistent configuration.

This does not mean that objects do not exist.

Objects are clearly observed.

They can be measured.

Their boundaries can often be distinguished.

Their motion can be tracked.

The point is simply that an observable object need not automatically be identical to the most fundamental unit of explanation.

What Is the Difference Between a Structure and an Object?

The words structure and object may appear almost interchangeable.

Within this chapter, however, we can use them with slightly different emphasis.

Structure emphasizes the condition in which multiple relationships form a particular configuration.

Object emphasizes the case in which that configuration is observed as if it were one distinguishable unit.

An object can therefore be understood as an observational expression of a structure.

Viewed from a great distance, a star may appear as a single point.

Viewed more closely, it reveals enormous internal dynamics.

At still smaller scales, more constituents and relationships appear.

The distinction of "one object" therefore depends partly on observational scale.

That does not make the object merely an illusion.

A sufficiently persistent and distinguishable configuration really does exist at that scale.

What Is a Momentum Structure?

In the previous section, we described structure as a persistent configuration of multiple coexisting Momentum.

We can now examine this expression more carefully.

Within one Island, there are many Differences.

Each Difference exists under different relational conditions.

Some relationships have short Distances.

Others have longer Distances.

Each relationship carries different Momentum.

Those Momentum may reinforce one another,

constrain one another,

redirect one another,

or partially counterbalance one another.

When the overall configuration remains within a sufficiently persistent range, a structure becomes observable.

A Momentum structure is therefore not a structure produced by one gigantic Momentum.

It is closer to:

a structured coexistence of many Momentum.

What matters is the configuration.

It is not enough to ask which Momentum exist.

We must also ask how those Momentum coexist within a relational network.

Does an Object "Have" Momentum?

We commonly say that an object has momentum.

Within conventional mechanics, this is a precise and useful expression.

From the relational perspective of DISTANCE, however, we can descend one level further.

Instead of assuming that an object exists first and Momentum is then assigned to it, we can consider whether multiple Momentum relationships form a configuration through which an object becomes observable.

Thus, beneath the observational expression:

object has Momentum

there may be a more relational description:

Momentum relations form an observable object.

These statements need not exclude one another.

The first belongs to the level of observation and calculation.

The second asks how the observed object itself can persist as a relational configuration.

Does One Object Contain Only One Momentum?

Clearly not.

Consider a star.

The star as a whole may move in a particular direction.

But within it, countless Momentum relationships exist.

There are binding relationships among its constituents.

There are transfers and transformations of energy.

There are internal dynamics.

There are also relationships with the external universe.

Describing the observable motion of the star using one momentum vector does not mean that every Momentum within the star has been reduced to that single quantity.

The same applies to stones, planets, and galaxies.

The observable motion of an object may be one expression of the directionality that becomes effective at the scale of the structure as a whole.

Beneath it, countless other Momentum coexist.

This returns us directly to a central proposition from Chapter 1:

The universe is therefore never composed of independent objects acted upon by independent Momentum. It is composed of countless coexisting Momentum whose continuously changing balance determines the observable directionality of every physical system.

We can now read this statement more concretely.

The object itself is not independent of that configuration of coexisting Momentum.

The Concept of an Island

The meaning of Island can now be made more precise.

An Island is not a perfectly isolated object.

It is a condition in which certain relationships are relatively more strongly effective, forming a configuration that persists long enough to become distinguishable from its surrounding relational network.

An Island is therefore:

a relatively persistent relational configuration.

Its persistence is not absolute.

Nor is its isolation absolute.

It is not even necessary that every internal relationship always be stronger than every external one.

What matters is sufficient relative coherence for one observable configuration to emerge.

Stars, planets, stones, and other objects can therefore be interpreted in DISTANCE as Islands.

Where Is the Boundary of an Island?

To call something an object seems to require some kind of boundary.

But we need not assume from the outset that this boundary must be a perfectly defined geometric surface.

Some relationships may be relatively more strongly effective within the configuration.

Relations with the surroundings may operate under different conditions.

This contrast can allow one configuration to become distinguishable from another.

The boundary may therefore first be understood as:

a relational transition.

At a particular scale, that relational transition may appear as a geometric boundary.

This distinction later becomes important in Chapter 4.

When we examine circles, spheres, centers, and physical boundaries, it prevents us from treating the boundary of an object as something that must already have been given in advance.

What Maintains the Identity of an Object?

If an object is a persistent configuration, the question of identity also changes.

Some of the constituents of an object may change while we continue to identify it as the same object.

This becomes especially obvious in living systems, but related problems appear in physical structures as well.

The identity of an object may therefore not depend solely on every constituent remaining exactly the same.

The persistence of a particular relational pattern, boundary, or configuration may be more important.

This opens the possibility that:

identity may persist through relational continuity even while constituents change.

This idea will become much more important later in the discussion of life.

For now, we remain at the level of physical objects.

An Object Is Not a Motionless Lump

Viewing an object as a Momentum structure also changes the meaning of rest.

Suppose a stone is resting on a desk.

It is clearly an object.

But its internal relationships have not all stopped.

Nor have its external relationships disappeared.

An object is therefore not a change-free entity.

It can instead be described as:

a configuration in which change is organized strongly enough for persistence to become observable.

Rest means only that the geometric position of the configuration as a whole changes very little within a particular observational frame.

It does not mean the absence of Momentum.

What Does It Mean for an Object to Move?

We now arrive naturally at the next question.

If an object is a Momentum structure, what exactly does it mean to say that the object moves?

Does a completed substance simply pass through empty space?

Or is the relational configuration that constitutes the structure continually readjusted within a changing relational context, with the resulting transformation appearing to us as motion?

This question is the bridge to Chapter 3.

So far, we have been asking how an object can appear as a persistent structure.

Next, we must reconsider the motion of that structure.

Observable Objects Depend on Scale

One further point matters.

Objects are related to observational scale.

Viewed from far away, Earth is one Island.

Standing on its surface, we distinguish countless separate structures.

A stone is one object at one scale and a complex relational network at another.

The number of objects we perceive is therefore not completely independent of the scale through which we observe them.

Again, DISTANCE is not trying to reduce objects to subjective illusion.

Persistent relational configurations really do exist at particular scales.

The point is simply that we need not assume that nature possesses only one absolute decomposition into objects.

Is It Enough to Keep Looking for Smaller Objects?

This line of thought leads to a deeper question.

A star can be decomposed into smaller constituents.

Those constituents can be decomposed further.

And those components may in turn reveal still smaller structures.

If we eventually discover the smallest fundamental object, will everything then be explained?

DISTANCE does not answer that question here.

But one possibility is already becoming visible.

Smaller size does not necessarily mean deeper explanation.

What may matter more is the set of relationships through which constituents form particular Momentum structures.

This question will return near the end of Chapter 4 when we examine light and the limits of physical boundaries.

For now, we do not take the argument further.

Is Structural Persistence a Form of Local Closure?

Once an Island forms, its internal relationships may appear partially closed.

But the closure is never complete.

External Momentum continue to exist.

Broader relational rearrangement continues.

An object is therefore not a completely independent system separated from wider Equalization.

It is better understood as:

local persistence within a continuing global relational process.

From this perspective, an object and the universe need not belong to completely different explanatory categories.

An object is a local configuration that persists within a larger relational network.

Stability Does Not Mean the Absence of Momentum

The longer an Island persists, the more stable it appears.

But stability does not imply that Momentum has disappeared.

Multiple Momentum may instead constrain one another in ways that sustain a particular configuration.

Some Momentum are resolved.

Others become newly effective.

Yet the macroscopic configuration remains within a certain range.

Stability can therefore be understood not as a static condition, but as relationally maintained persistence.

This becomes important in Chapter 4 when DISTANCE reinterprets mass, gravity, inertia, and rotation.

Objects and Equalization

We can now connect objects directly to Equalization.

An object is not what remains after Equalization has ended.

Nor is it an exception that obstructs Equalization.

It can emerge through a process such as:

Momentum resolution → relational rearrangement → local Momentum reconstruction → persistent configuration

The existence of objects therefore does not contradict Equalization.

It instead reinforces the idea that Equalization is not simple homogenization, but continuing relational reconfiguration.

More Objects Do Not Necessarily Mean Simpler Relationships

The early universe may not have contained objects of the kinds we identify today.

The present universe contains countless distinguishable Islands.

Does this mean relationships have simply become separated into simpler units?

Not necessarily.

As global Momentum was resolved, the variety and hierarchy of local relationships may have increased.

One extremely dense relational configuration may have been reorganized into countless nested Islands.

Cosmic history may therefore be understood not merely as fragmentation, but as an increasing differentiation of relational configurations.

This becomes an important foundation for understanding the later emergence of life.

From Object to Structure, and from Structure to Motion

The flow of Chapter 2 can now be seen more clearly.

The early universe existed in an extremely strong relational state.

That state contained enormous unresolved Momentum.

As Momentum was resolved through Equalization, cosmic diffusion proceeded.

The relational network was continuously rearranged.

Within that process, new local Differences and Momentum became effective.

Some Momentum configurations persisted for relatively long intervals.

Those persistent configurations are structures.

And when an observer recognizes such a structure as one distinguishable unit, it appears as an object.

Thus, provisionally:

object = observable persistent Momentum structure

And these objects, in DISTANCE, are also Islands.

Conclusion of Chapter 2

The post-Big-Bang universe need not be understood only as a collection of objects scattered through an expanding space.

DISTANCE allows another picture.

There was an extremely dense relational network.

Strong Momentum within it were resolved.

Relational Distances changed.

The network was rearranged.

New local relationships emerged.

Some configurations persisted as structures.

The stars, planets, galaxies, stones, and other objects we observe today can be understood as some of the outcomes of that process.

The observable universe can therefore be read not simply as a collection of completed independent objects, but as:

a relational field of temporarily persistent Momentum structures.

Toward the Next Chapter — What Does It Mean for a Structure to Move?

The direction of the question now changes.

Chapter 2 asked:

How can structure emerge?

The next chapter must ask:

What does it mean for that structure to move?

When one Island moves from one position to another, what exactly has changed?

Has a completed structure simply been transported intact through space?

Or are the relationships constituting that structure and the surrounding relational network being continually rearranged, with the result appearing as motion?

And how do local Momentum and broader Momentum coexist within that process?

These questions provide the starting point for the next chapter:

Chapter 3 — The Relational Structure of Motion

We now move one level beneath the familiar picture of objects moving through the universe and ask how changes in Momentum structures become observable as motion.